

One potentially significant yet little investigated criterion for postoccupancy evaluation is the "legibility" of a setting—the degree to which a building facilitates the ability of users to find their way within it. The present study evaluated the legibility of a sample of ten university buildings. Self-report data indicated way-finding to be a problem for a substantial minority. The impact upon way-finding of several theoretically derived visual/spatial variables was also assessed. Two aspects of the floor plan configurations of these settings, as judged from highly significant relationships to frequency of disorientation. One of these variables, judged simplicity of floor plan configuration, was able to account for 56% of the variance in these data. One other potentially important variable, respondents' own familiarity with these buildings, was able to account for but 9% of the variance in frequency of disorientation data.

EVALUATING ARCHITECTURAL LEGIBILITY

Way-Finding in the Built Environment

JERRY WEISMAN *is Assistant Professor in the Program in Man-Environment Relations at the Pennsylvania State University. His teaching and research interests focus on behavioral programming and evaluation for environmental design; environmental cognition and way-finding behavior; and the planning, programming, and management of facilities for the elderly. He holds a Ph.D. in Psychology & Architecture from the University of Michigan.*

The ability to effectively find one's way into, through, and out of a building is clearly a prerequisite for the satisfaction of other, higher level goals. Thus the "legibility" of an environment—the extent to which it facilitates the process of way-finding—may have significant behavioral consequences and, as Friedman et al. (1978) have argued, should be con-

AUTHOR'S NOTE: *My thanks to Jan Reizenstein and Craig Zimring, Gary Evans, and Rachel and Stephen Kaplan for their comments on earlier versions of this article.*

sidered as a potentially important criterion in postoccupancy evaluations.

Relatively few evaluations, however, have incorporated way-finding behavior and environmental legibility into their design. While the housing studies of Zeisel and Griffin (1975) and Cooper (1975) give some attention to issues of orientation and spatial confusion, their focus is outdoor rather than interior spaces. Berkeley (1973), Dixon (1968), and McKean (1972) report that anger, hostility, and indignation resulted when users were faced with "illegible" public buildings; these respective evaluations were, however, all quite informal in nature. More extensive studies of way-finding behavior within buildings have been conducted by Best (1970), Corlett et al. (1972), and Evans (1979), albeit not within the context of comprehensive environmental evaluations.

Thus, only very limited empirical data are available regarding the nature or pervasiveness of way-finding problems within the built environment. Furthermore, since few of the extant studies consider the specific environmental context within which way-finding occurs, the ways in which visual/spatial features of architectural settings may contribute to, or help to resolve, problems of way-finding are little understood. It should also be recognized that way-finding is a complex cognitive task; yet only a few studies (e.g., Evans, 1979) take an explicit theoretical position regarding underlying psychological structures and processes.

The present study, therefore, is structured in three parts: the first focuses upon the formulation and measurement across a sample of 10 settings of theoretically derived environmental variables seen as relevant to way-finding; the second concerns the gathering of self-report data on way-finding from study participants familiar with many or all of these same sample settings; the third assesses the influence of these environmental variables upon participants' reported way-finding behavior.

ENVIRONMENTAL VARIABLES AND SAMPLE SETTINGS

DEFINING ENVIRONMENTAL VARIABLES

The definition of relevant variables in environmental research is often the result of testing post hoc inferences made once data have been gathered. The present study, however, took a quite different approach. Building upon theoretical considerations, specifically the model of environmental cognition advanced by S. Kaplan (1973, 1976), four broad categories of variables were defined at the outset of the study. These variables emerge from the view of way-finding as a cognitive task. It is assumed that most architectural settings, as with larger scale environments, are too extensive to be perceived in their entirety from any one location; it is necessary that information regarding specific locations, and the spatial relationships among locations, be stored in one's head. Thus it was hypothesized that such visual/spatial variables as type(s) of signage provided, the ability to see through or out of a setting, the extent to which one location looks "different" from others, and the overall plan or layout of a setting may all impact way-finding behavior. It is the fourth of these variables, plan configuration, which will be the focus of the present discussion.

It is essential to recognize that each of these classes of environmental variables may be manifested in different ways in different settings. Indeed R. Kaplan has suggested that "the portion of the environment under examination inevitably turns out to be a *construct*, a multifaceted entity with unclear boundaries" (1977: 248). Thus it is essential to examine such variables, and their behavioral consequences, not just in one setting, as in the typical case study approach with its limitations on generalizability (Sommer, 1977: 201), but across a *sample* of settings (R. Kaplan, 1975).

SAMPLE SETTINGS

Ten buildings on the Central Campus of the University of Michigan comprise the environmental sample for this study. Pretesting indicated these to be frequently utilized settings, relatively well-known to most students. These settings were likewise seen as possessing some reasonable degree of variability with respect to the four broad classes of visual/spatial features outlined above.

Eight of the 10 buildings serve mixed classroom/office/laboratory functions; one is essentially an office block, and another houses administrative as well as academic offices. Eight of the 10 buildings are either three or four stories in height; the remaining two are towers 8 and 11 stories high. The newest of the buildings was completed in the early 1970s; the oldest date to the early years of this century.

The 10 sample settings vary considerably in both size and configuration of their floor plans. The smallest is a simple "I" shape, under 200 feet in length; the longest is well over 500 feet. In addition to simple "I" plans, there are "L", "T", and "V" shaped buildings as well as three settings whose plans might be characterized as hollow squares. All buildings have predominantly "double-loaded" corridors, with rooms arrayed along both sides. Floor plans of two of the sample settings are displayed in Figure 1.

PLAN CONFIGURATION

As suggested above, the overall plan configuration of a building, and particularly the ease and accuracy with which one can build a mental image of it, may have some considerable impact upon way-finding behavior. Thus McKean, in discussing the proliferation of floor plans pinned to the walls within the University of Essex, suggests that they are "a reflection of the incomprehensibility of the complex" (1972: 664). Several environmental researchers (Appleyard, 1970; DeJonge, 1962; Lynch, 1960) have suggested much

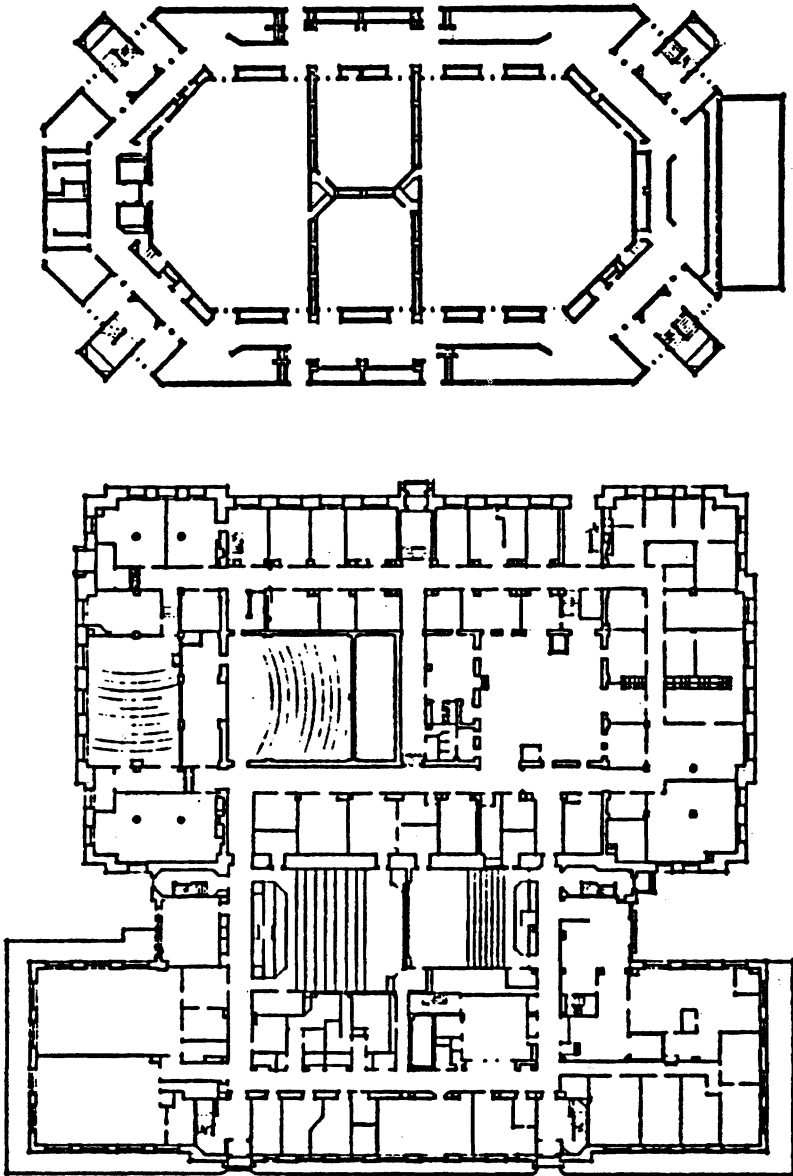


Figure 1: Two of the 10 study settings, the Modern Languages and Chemistry Buildings. Floor plans are drawn to the same scale.

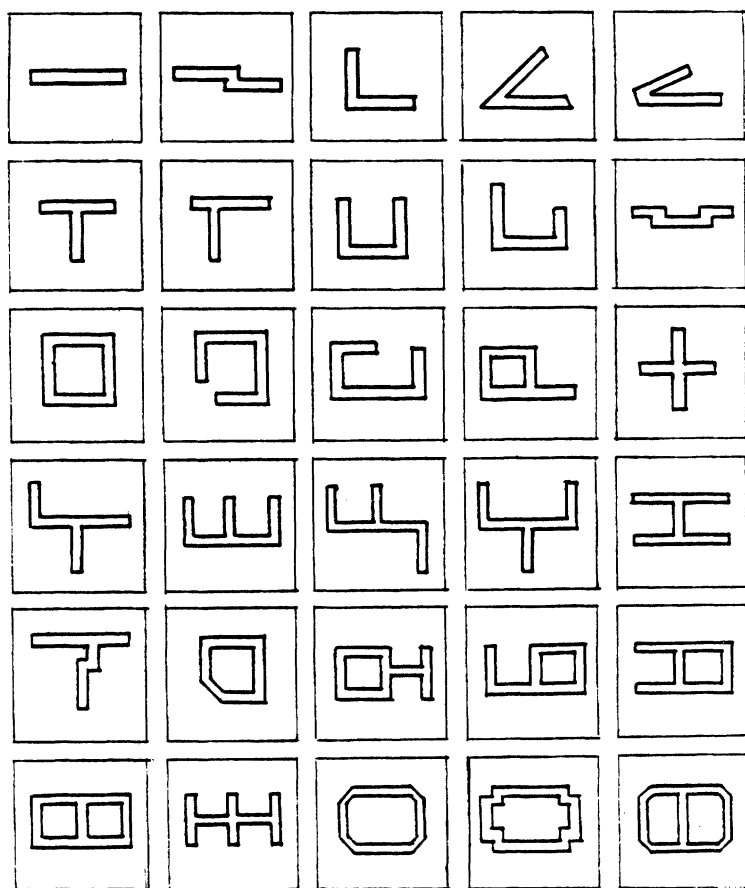


Figure 2: Set of 30 diagrams utilized to assess "good form" of plan configurations.

the same with respect to urban scale settings. Specifically, they have invoked the Gestalt psychology laws of perceptual organization and "good form" in characterizing spatial relationships most readily represented in an individual's cognitive map. Among those qualities of "good form" seen as relevant are symmetry, regularity, and continuity (Canter, 1974: 32-34).

ASSESSMENT OF "GOODNESS OF FORM"

To assess the good form characteristics of the sample settings under study, a set of 30 highly simplified and abstracted diagrams were developed (see Figure 2). Twenty of these diagrams are simplifications of the plans of actual University of Michigan buildings, including the 10 sample settings used in this study. The remaining 10 diagrams represent additional variations of plan configuration constructed on the basis of manipulation of number of corridor segments, overall symmetry, and treatment of corner intersections.

A variety of judgments were then made by five groups of independent raters with respect to this set of diagrams. Each group of 20 raters provided judgments with respect to *one* of the following criteria: (1) *preference* for each diagram; (2) the *complexity* of each diagram; (3) the ease with which each diagram could be *described to another*; (4) the ease with which one could *memorize* each diagram; and (5) the *judged ease of way-finding* in a building with such a plan configuration. The first four of these criteria emerged from previous research (Alexander and Carey, 1968), which indicated strong relationships among patterns with these characteristics. The use of the five groups of independent raters follows the methodological considerations proposed by R. Kaplan (1975).

It should be emphasized that the raters responding to the first four of these criteria were presented with the 30 diagrams as nothing more than abstract two-dimensional stimuli; they were given no indication that the diagrams represented abstractions of actual building plans. Similarly, no indication was given to the final group of raters that 20 of the plans they were evaluating were derived from actual settings on the Michigan campus; no raters indicated that they were aware of this relationship.

Based upon the mean ratings for each of the judgments, the 10 study settings were dichotomized into "high" and "low" groupings with respect to each aspect of good form of plan configuration.

WAY-FINDING BEHAVIOR IN THE STUDY SETTINGS

PARTICIPANTS

Independent of the assessment of plan configuration reported above, self-report data on perceptions of and way-finding behavior in the 10 study settings were gathered from a wholly different group of participants.

There were 73 University of Michigan undergraduates who participated in this segment of the study. This sample was about equally divided among men and women. Approximately 25% were first-year students, 60% were second-year students, and the remainder in their final two years. At the time the data were gathered, the newest of the students had been on campus a minimum of six weeks to two months.

These participants were "tested" in small groups of 4 to 12. All were enrolled in classes in the Psychology Department and were fulfilling a course requirement for participation in an experimental study.

QUESTIONNAIRE

A questionnaire format was utilized to gather self-report data on participants' way-finding behavior in, and perceptions of, the 10 study settings. Participants responded to a set of 10 questions for each of the 10 buildings (see Figure 3). The order of presentation of the buildings was reversed for one half of the participants to control for problems of fatigue and/or boredom which might have resulted from repeated response to the same set of questions. The questionnaire focused on three sets of issues: familiarity with

-
- 1) How often have you had occasion to use - to be inside - this building?
 1 never* 2 once or a few times 3 fairly often 4 regularly
- (*If you have never been inside this building, simply go on to the next page)
- 2) How long has it been since the last time you were inside of this building?
 1 a year or more 2 a month or more 3 a week or more 4 a week or less
- 3) Roughly what percentage of the total building would you say you've been in?
 1 25% or less 2 25% to half 3 half to 75% 4 all or almost all
- 4) Have you ever had trouble finding your way or gotten 'lost' inside of this building? How often has this happened?
 1 = just about every time I walk in there
 2 = fairly often
 3 = just occasionally
 4 = practically never (if 'practically never' go on to Question 6)
- 5) What was the 'most lost' you've ever become in this building?
 1 totally lost 2 _____ 3 _____ 4 just momentarily disoriented
- 6) Do you think you'd be able to direct a stranger to many places (eg. classrooms, stairways, exits) inside this building? How many places?
 1 practically nowhere 2 _____ 3 _____ 4 most anyplace
- 7) How confident would you be of the directions you'd give to such a stranger?
 1 not at all confident 2 _____ 3 _____ 4 quite sure
- 8) All things considered, do you find this building a relatively easy or a relatively difficult place to 'figure out' and in which to find your way?
 1 difficult 2 _____ 3 _____ 4 easy
- 9) Do you think you could draw a quick diagram of this building's 'layout' - showing the outline of the building and the arrangement of the corridors inside (you don't have to do the actual sketch)? Would doing such a diagram of this building be difficult or easy? (Don't worry about your own degree of 'artistic' ability)
 1 difficult 2 _____ 3 _____ 4 easy
- 10) How confident would you be of the building sketch you'd have drawn?
 1 not confident 2 _____ 3 _____ 4 quite sure
-

Figure 3: Set of questions responded to for each of the 10 study settings.

each setting; reported way-finding behavior within each setting; and perceived knowledge and understanding of each setting.

The present analyses focus on the first two of these three sets of issues. Questions 1 to 3 tap frequency, recency, and

extent of usage as measures of familiarity with each building. Questions 4 and 5 ask respondents to report the frequency and extent of their disorientation within each building. The same response format, a four-point scale, is utilized for all questions for all buildings.

WAY-FINDING BEHAVIOR, FAMILIARITY, AND ENVIRONMENTAL VARIABLES

Three interrelated questions will be considered in discussing the findings of this study: (1) the nature and pervasiveness of participants' way-finding problems in the 10 sample settings; (2) the relationship between way-finding behavior and participants' familiarity with these settings; and (3) the relationship between way-finding behavior and various aspects of the good form of the plan configurations of these settings.

WAY-FINDING BEHAVIOR

"On the average," way-finding behavior was not reported to be a substantial problem in any of the 10 campus buildings evaluated. For the 10 settings, mean scores for frequency of successful way-finding (Question 4) ranged from a high of 3.68 (on a 4-point scale) to a low of 2.97. Mean scores for extent of successful way-finding (Question 5) present an only slightly greater range, with means from a high of 3.48 to a low of 2.40.

Inspection of frequency distributions, for Questions 4 and 5, however, reveals a rather different situation. While the majority of study participants seemingly experience little or no difficulty in way-finding within the study settings, problems in orientation are reported by a sizable minority of students. Averaging across all 10 buildings, 9.1% of respondents at least minimally familiar with these settings report becoming lost often or almost always; 14.1% have been, on at least one occasion, quite or totally lost, while both

frequent and severe disorientation are reported by 8% of respondents.

The percentages of users reporting themselves lost often or virtually always varies from 0% to 6% in five of the buildings and from 10% to 26% in the other five. With respect to the extent of disorientation, percentages range even higher. Almost 40% of Chemistry Building users report having become quite or totally lost; 23% of Frieze Building users indicate a similar extent of disorientation, as do 18% of those familiar with the Natural Sciences Building, and 13% of Modern Languages Building users. The data also indicate a strong relationship between the two aspects of way-finding, with those reporting frequent or constant disorientation having been quite or totally lost as well.

WAY-FINDING AND FAMILIARITY

One obvious and potentially powerful influence upon way-finding behavior may be the degree of familiarity an individual has with a given setting. If increased familiarity is sufficient to overcome any initial difficulties in orientation, then efforts might simply be directed toward increasing the knowledge level of naive users of a setting. If, to the contrary, familiarity alone does not explain disorientation, then other factors, such as visual/spatial features of the environment, ought to be considered.

The study data do not suggest any particularly simple relationship between reported way-finding behavior and any of the facets of familiarity tapped in the questionnaire (i.e., frequency, recency, or extent of usage). Correlational analyses, based upon the 10 sample settings as cases, yielded no significant relationships.

Correlations between reported way-finding behavior and each of the three facets of familiarity were also computed for individual buildings. While these yielded significant relationships for 5 of the 10 buildings, in only three cases was familiarity able to account for more than 10% of the variance in way-finding behavior; in no case was familiarity

TABLE 1
Results of Paired t Tests for Environmental Variables^a

	MEAN		t	p
	LOW	HIGH		
PLAN CONFIGURATION				
PREFERENCE	3.42	3.30	1.80	NS
SIMPLICITY	3.15	3.58 ^b	7.03	.001
DESCRIBABILITY	3.27	3.43 ^c	2.27	.05
MEMORABILITY	3.27	3.42 ^c	.88	NS
PLAN LEGIBILITY	3.19	3.55	6.70	.001

a. n's for the five tests varied from a low of 58 to a high of 65.

b. Means are for Complex and Simple diagrams, respectively.

c. Means are for Difficult and Easy diagrams, respectively.

able to account for more than 25% of the variance. In some settings, such as the Modern Languages Building, the majority of those users reported as often/always lost actually had moderate or high levels of familiarity. The opposite situation prevailed in the Natural Sciences Building where, among those lost often/always, virtually no one had a moderate or high level of familiarity. In some environments continuing experience and increased familiarity seemingly come to mitigate way-finding problems; in other settings, however, more frequent and extensive usage may simply provide an increased "opportunity" to become lost.

PLAN CONFIGURATION AND WAY-FINDING BEHAVIOR

As described earlier, the 10 sample settings were dichotomized into high and low groupings with respect to each of the various aspects of good form of plan configuration. These groupings were then utilized as the basis for a series of correlated or pair-wise t tests. Taking only study participants familiar with at least three of the five buildings in

each grouping (e.g., high and low complexity), mean frequency of disorientation scores was computed for each individual for the high and low groups. These two scores per subject were then utilized to test the null hypothesis that the mean of the differences in way-finding scores between high and low groupings would be zero (Blalock, 1972: 233-235). It was hypothesized that reported way-finding scores would be significantly better for those buildings rated as "high" with respect to judgments of simplicity, preference, describability, memorability, and ease of way-finding for the abstracted plan configuration diagrams.

Results of the correlated *t* tests are displayed in Table 1. Highly significant and expected differences were found for two aspects of plan configuration, judged simplicity and judged ease of way-finding.

Correlational analyses were also undertaken to further explore these relationships. A strong and significant relationship was found between judged simplicity of plan configuration diagrams and reported frequency of disorientation across the 10 buildings ($r = .75, p \leq .05$). A strong but less than significant relationship was found between reported way-finding and ease of way-finding as judged from the abstract plan diagrams.

CONCLUSIONS, LIMITATIONS, AND IMPLICATIONS

Evaluation of 10 sample settings on the University of Michigan campus indicates that successful way-finding is a problem for a substantial minority of users of many of these buildings. Furthermore, aspects of floor plan configurations, as judged from highly abstracted two-dimensional diagrams, are strong predictors of reported way-finding behavior. In particular, simplicity judgments, provided by a group of independent raters, were able to account for 56% of the variance in reported frequency of disorientation data. By comparison, participants' own familiarity with the study settings (specifically recency of usage) was able to account for but 9% of the variance in these way-finding scores.

Way-finding behavior as assessed in the present study is, of course, based upon self-report data provided by study participants; questions might thus be raised regarding the relationship of such self-report data to more objective measures of way-finding behavior. While the limitations of self-report data must be recognized, at least one recent study (Kozlowski and Bryant, 1977) indicates a positive relationship between self-estimation of respondents' "sense of direction" and their actual way-finding performance on several tasks.

Given the predictive power of judgments of plan configuration diagrams as utilized in this study, such simple paper-and-pencil tests may be of value for both postoccupancy evaluation and for what might be characterized as "preconstruction simulation." There has, in recent years, been some considerable interest in the presentation of surrogates of environmental experience to assess behavioral responses to as yet unbuilt design and planning alternatives (Appleyard et al., 1973; S. Kaplan, 1977). The plan configuration diagrams described above, perhaps combined with various forms of photographic or video presentation, should allow for simulation and evaluation of a variety of visual/spatial design variables potentially related to way-finding behavior.

Finally, while the relationship between at least one aspect of architectural legibility and way-finding behavior may now be more clearly drawn, other behavioral consequences of illegible settings remain to be explored. Lawton (1975: 161) suggests that the clarity or legibility of the physical environment may impact the "successful adjustment" of elderly residents of congregate housing. In their discussion of therapeutic environments, Canter and Canter elaborate on this issue:

In order to comprehend an organization and take advantage of it, it is necessary to understand how it is arranged in space [1979: 6].

Indeed, Lynch (1960: 5) has suggested that "a distinctive and legible environment . . . heightens the potential depth

and intensity of human experience." It remains for such consequences of legibility to be evaluated.

REFERENCES

- ALEXANDER, C. and S. CAREY (1968) "Subsymmetries." *Perception and Psychophysics* 4: 73-77.
- APPLEYARD, D. (1970) "Styles and methods of structuring a city." *Environment and Behavior* 2: 100-117.
- K. CRAIK, M. KLAPP, and A. KREIMER (1973) *The Berkeley environmental simulation laboratory: Its use in environmental impact assessment.* Working Paper No. 206. Berkeley: University of California, Institute of Urban and Regional Development.
- BERKELEY, E. (1973) "More than you wanted to know about the Boston City Hall." *Architecture Plus* (February): 72-77.
- BEST, G. (1970) "Direction-finding in large buildings," pp. 72-75 in D. Canter (ed.) *Architectural Psychology*. London: RIBA Publications.
- BLALOCK, H. (1972) *Social Statistics*. New York: McGraw-Hill.
- CANTER, D. (1974) *Psychology for architects*. New York: Wiley.
- and S. CANTER. (1979) *Designing for Therapeutic Environments: A Review of Research*. New York: Wiley.
- COOPER, C. (1975) *Easter Hill Village: Some Social Implications of Design*. New York: Free Press.
- CORLETT, E., I. MANENICA and R. BISHOP (1972) "The design of direction-finding systems in buildings." *Applied Ergonomics* 3: 66-69.
- DE JONGE, D. (1962) "Images of urban areas: Their structure and psychological foundations." *J. of the Amer. Institute of Planners* 28: 266-276.
- DIXON, J. (1968) "Campus city revisited." *Architectural Forum* 129 (5): 28-43.
- EVANS, G., J. FELLOWS, M. ZORN, and K. DOTY (1980) "Cognitive mapping and architecture." *J. of Applied Psychology* 65: 474-478.
- FRIEDMAN, A., C. ZIMRING, and E. ZUBE (1978) *Environmental Design Evaluation*. New York: Plenum.
- KAPLAN, R. (1977) "Preference and everyday nature: Method and application," pp. 235-250 in D. Stokols (ed.) *Perspectives on Environment and Behavior: Theory, Research and Application*. New York: Plenum.
- (1975) "Some methods and strategies in the prediction of preference," pp. 118-129 in E. Zube et al. (eds.) *Landscape assessment: Values, Perceptions and Resources*. Stroudsburg, PA: Dowden, Hutchinson and Ross.
- KAPLAN, S. (1977) "Participation in the design process: A cognitive approach," pp. 221-233 in D. Stokols (ed.) *Perspectives on Environment and Behavior: Theory, Research and Application*. New York: Plenum.
- (1976) "Adaptation, structure and knowledge," pp. 32-45 in G. Moore and R. Golledge (eds.) *Environmental Knowing: Theories, Research and Methods*. Stroudsburg, PA: Dowden, Hutchinson and Ross.
- (1973) "Cognitive maps in perception and thought," pp. 63-78 in R. Downs and D. Stea (eds.) *Image and Environment: Cognitive Mapping and Spatial Behavior*. Chicago: Aldine.

- KOZLOWSKI, L. and K. BRYANT (1977) "Sense of direction, spatial orientation and cognitive maps." *J. of Experimental Psychology: Human Perception and Performance* 3: 590-598.
- LAWTON, M. P. (1975) *Planning and Managing Housing for the Elderly*. New York: Wiley-Interscience.
- LYNCH, K. (1960) *Image of the City*. Cambridge: MIT Press.
- McKEAN, J. (1972) "University of Essex: Case study." *Architects J.* 156: 645-667.
- SOMMER, R. (1977) "Action research," pp. 195-203 in D. Stokols (ed.) *Perspectives on Environment and Behavior: Theory, Research and Applications*. New York: Plenum.
- ZEISEL, J. and M. GRIFFIN (1975) *Charlesview Housing: A Diagnostic Evaluation*. Cambridge, MA: Architecture Research Office, Graduate School of Design, Harvard University.